

Corporation of the City of Quinte West

Quinte West Drinking Water System (Bayside Service Area)

2024 Annual & Summary Drinking Water System Report





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Public Works and Environmental Services
Water/Wastewater Division
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Drinking Water System Information

Drinking Water System Number:	220008079
Drinking Water System Name:	Quinte West Drinking Water System (Bayside Service Area)
Drinking Water System Owner:	The Corporation of the City of Quinte West
Drinking Water System Category:	Large Municipal-Residential System
Period being Reported:	January 1, 2024 through December 31, 2024

Does your Drinking Water System service more than 10,000 people?

Yes.

Is your Annual Report available to the public at no charge on a website

Yes, please visit www.quintewest.ca

Location where Summary Report required under O.Reg.170/03 Schedule 22 will be available for inspection

Water/Wastewater Services Administration Office

25 Couch Crescent

Trenton, ON, K8V 1G8

List all Drinking Water Systems (if any), that receive all of their Drinking Water from your System:

None.



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Indicate how you notified system users that your annual report is available, and is free of charge.

- ☒ Public access/notice via City website
- ☐ Public access/notice via Government Office
- ☐ Public access/notice via newspaper
- ☒ Public access/notice via City social media platform(s)
- ☐ Public access/notice via a Public Library
- ☐ Public access/notice via other method:

Description of the Drinking Water System

The Bayside Water Treatment Plant pulls water from the Bay of Quinte through a 404 m long intake pipe that extends approximately 370 m into the Bay. This conventional, chemically-assisted filtration plant has a rated capacity of 11,360 cu.m/day. The facility uses rapid-mix coagulation, flocculation, sedimentation, dual-media filtration, and granular activated adsorber cells. The facility uses sodium hypochlorite for disinfection, and fluoridates its drinking water supply. Drinking Water is pumped to the Water Distribution System (WDS) via two (2) vertical turbine highlift pumps that draw water from the onsite 980 cu.m. clearwell. The elevated water storage tower located at 10 Richards Way, has an effective usable volume of 6,050 cu.m. A dual-celled holding pond is used to provide natural filtration for process wastewater prior to final discharge into an adjacent wetland through ex-filtration.

The Bayside Water Treatment Plant services approximately 5,000 customers in the community of Bayside, in addition to a portion of CFB Trenton. In 2015, the City constructed a watermain that effectively connected the Bayside WDS to the Trenton WDS. On June 20, 2023, the Trenton/Bayside Drinking Water Systems (DWS) and Frankford/Batawa DWS, previously governed by their own Drinking Water Works Permit (DWWP) and Municipal Drinking Water Licence (MDWL), were consolidated under one



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DWWP # 163-202, and one MDWL # 163-102, the Quinte West DWS. Each system is still assigned a separate DWS number by the Ministry of Environment, Conservation and Parks (MECP).

List all water treatment chemical used over this Reporting Period:

- ✓ Polyaluminum Chloride (Tradename: SternPac)
- ✓ Sodium Hypochlorite
- ✓ Hydrofluosilicic Acid (Fluoride solution)

Were any significant expenses incurred to:

- ✓ Install Required Equipment?
- ✓ Repair required Equipment?
- ✓ Replace required Equipment?

Provide a brief description, and a breakdown of monetary expenses incurred to facilitate Equipment upgrades:

Preventative Maintenance (PM) activities for lubrication, inspections, testing and cleaning of equipment is scheduled and completed routinely, along with other lifecycle replacement needs. In addition to the PM activity, the following Capital expenditures were incurred this Reporting Period:

Intake Inspections	~\$152,000 (combined with all DWS)
Customer Water Meter Automation and Replacements	~\$120,000 (combined with all DWS)
Highlift Pump Upgrade (including new motor and VFD)	~\$117,000



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Building Upgrades	~\$68,000
Reactivated Carbon	~\$35,000
Building Security Upgrades	~16,000
Overhaul Lowlift Pump #2	~\$12,000
Generator Controller Replacement	~\$11,000
Spill containment tank sealing	~\$5,000

Notices submitted in accordance with subsection 18(1) of the Safe Drinking Water Act or section 16-4 of Schedule 16 of O.Reg. 170/03 and reported to Spills Action Centre during this Reporting Period:

Incident Date (DD-MMM-YY)	Parameter	Result	Corrective Action	Corrective Action Date (DD-MMM-YY)
There were no notices submitted in accordance with subsection 18(1) of the SDWA or section 16-4 of Schedule 16 of O. Reg. 170/03 during this Reporting Period.				



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Microbiological Testing completed in accordance with Schedule 10, 11, or 12 of Regulation 170/03 during this Reporting Period:

	Number of Samples	Range of E.Coli (min - max)	Range of Total Coliform (min - max)	Number of Heterotrophic Plate Count Samples	Range of HPC results (min - max)
Raw	53	0 - NDOGT ¹	0 - NDOGT	NA	NA
Treated	53	0 - 0	0 - 0	53	0 - 16
Distribution	182	0 - 0	0 - 0	82	0 - 7

Operational Testing completed in accordance with Schedule 7, 8, or 9 of O.Reg.170/03 during this Reporting Period:

Parameter	Number of Grab Samples	Range of Results (min-max)
Turbidity, (NTU)	8760	0.03 - 0.66
Primary Disinfection FAC ² , (mg/L)	8760	1.24 ³ - 4.02 ⁴
Secondary Disinfection FAC, (mg/L)	8760	0.57 - 3.01
Fluoride, (mg/L)	8760	0.20 - 1.00 ⁵

¹ NDOGT stands for No Data: Overgrown with Target Bacteria. Prior, and subsequent datasets within normal range.

² Free Available Chlorine. Defined as the free amount of chlorine available in water.

³ Isolated incident. Highlift pump shut off.

⁴ Isolated incident. Suspected analyzer error, normal values below and after instantaneous spike.

⁵ Value recorded during highlift pump switch over.



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Summary of additional testing and sampling carried out in accordance with the requirement of an Approval, Order, or other Legal Instrument:

In accordance with Condition 5.4 of MDWL #163-102, issue number 6, monthly samples were collected from the 'Filter to Waste' environmental Discharge and tested for Total Suspended Solids.

Condition 1.5 of the MDWL outlines the maximum annual average concentration for each of these parameters. Results of this sampling and testing are outlined in the table below.

Residue Management, TSS Results, (mg/L)	
Date	Filter to Waste Discharge
January 3, 2024	< 2
February 6, 2024	< 2
March 5, 2024	< 2
April 3, 2024	< 2
May 7, 2024	< 2
June 4, 2024	2
July 3, 2024	< 2
August 7, 2024	< 2
September 4, 2024	< 2
October 2, 2024	3
November 5, 2024	2
December 3, 2024	2
Annual Average Concentration (mg/L)	2



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**Summary of Inorganic parameters tested during this Reporting Period
 or the most recent Sample Results:**

Inorganic Compound Results				
Parameter	Sample Date (dd/mm/yy)	Result Value	Unit of Measure	Exceedance?
Antimony	03-JAN-24	< 0.6	ug/L	No
Arsenic	03-JAN-24	0.2	ug/L	No
Barium	03-JAN-24	31.7	ug/L	No
Boron	03-JAN-24	10	ug/L	No
Cadmium	03-JAN-24	< 0.003	ug/L	No
Chromium	03-JAN-24	0.11	ug/L	No
Mercury	03-JAN-24	< 0.01	ug/L	No
Selenium	03-JAN-24	< 0.04	ug/L	No
Sodium	03-JAN-24	14.0	mg/L	No
Fluoride	03-JAN-24	0.49	mg/L	No
Uranium	03-JAN-24	0.043	ug/L	No
Nitrate	03-JAN-24 03-APR-24 03-JUL-24 02-OCT-24	0.295 0.253 0.153 0.115	mg/L	No
Nitrite	03-JAN-24 03-APR-24 03-JUL-24 02-OCT-24	0.003 0.003 0.003 0.003	mg/L	No



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Summary of Lead Testing under Schedule 15.1 during this Reporting Period:

Lead Sampling Results				
Sampling Period	Location Type	Number of Samples	Range of Lead Results (ug/L), 'min-max'	Number of Exceedances
Winter Sampling Period ⁶	Hydrant	4	0.10 - 0.46	None
Summer Sampling Period ⁷	Hydrant	4	0.01 - 0.04	None

Summary of Organic parameters sampled during this Reporting Period:

Organic Compound Results				
Parameter	Sample Date	Result Value	Unit of Measure	Exceedance
Alachlor	03-JAN-24	< 0.02	ug/L	No
Atrazine + N-dealkylated metabolites	03-JAN-24	< 0.01	ug/L	No
Azinphos-methyl	03-JAN-24	< 0.05	ug/L	No
Benzene	03-JAN-24	< 0.32	ug/L	No

⁶ Winter Sampling Period runs from December 15 - April 15.

⁷ Summer Sampling Period runs from June 15 - October 15.



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Benzo(a)pyrene	03-JAN-24	< 0.004	ug/L	No
Bromoxynil	03-JAN-24	< 0.33	ug/L	No
Carbaryl	03-JAN-24	< 0.05	ug/L	No
Carbofuran	03-JAN-24	< 0.01	ug/L	No
Carbon Tetrachloride	03-JAN-24	< 0.17	ug/L	No
Chlorpyrifos	03-JAN-24	< 0.02	ug/L	No
Diazinon	03-JAN-24	< 0.02	ug/L	No
Dicamba	03-JAN-24	< 0.20	ug/L	No
1,2-Dichlorobenzene	03-JAN-24	< 0.41	ug/L	No
1,4-Dichlorobenzene	03-JAN-24	< 0.36	ug/L	No
1,2-Dichloroethane	03-JAN-24	< 0.35	ug/L	No
1,1-Dichloroethylene (vinylidene chloride)	03-JAN-24	< 0.33	ug/L	No
Dichloromethane	03-JAN-24	< 0.35	ug/L	No
2-4 Dichlorophenol	03-JAN-24	< 0.15	ug/L	No
2,4-Dichlorophenoxy acetic acid (2,4-D)	03-JAN-24	< 0.19	ug/L	No



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Diclofop-methyl	03-JAN-24	< 0.40	ug/L	No
Dimethoate	03-JAN-24	< 0.06	ug/L	No
Diquat	03-JAN-24	< 1	ug/L	No
Diuron	03-JAN-24	< 0.03	ug/L	No
Glyphosate	03-JAN-24	< 1	ug/L	No
Malathion	03-JAN-24	< 0.02	ug/L	No
2 methyl-4-chlorophenoxyacetic acid (MCPA)	03-JAN-24	< 0.00012	ug/L	No
Metolachlor	03-JAN-24	< 0.01	ug/L	No
Metribuzin	03-JAN-24	< 0.02	ug/L	No
Monochlorobenzene	03-JAN-24	< 0.3	ug/L	No
Paraquat	03-JAN-24	< 1	ug/L	No
Pentachlorophenol	03-JAN-24	< 0.15	ug/L	No
Phorate	03-JAN-24	< 0.01	ug/L	No
Picloram	03-JAN-24	< 1	ug/L	No
Polychlorinated Biphenyls(PCB)	03-JAN-24	< 0.04	ug/L	No



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Prometryne	03-JAN-24	< 0.03	ug/L	No
Simazine	03-JAN-24	< 0.01	ug/L	No
Trihalomethanes (THMs) (Running annual average)	02-OCT-24	95.00	ug/L	No
Haloacetic acids (HAAs) (Running annual average)	02-OCT-24	79.63	ug/L	No
Terbufos	03-JAN-24	< 0.01	ug/L	No
Tetrachloroethylene	03-JAN-24	< 0.35	ug/L	No
2,3,4,6-Tetrachlorophenol	03-JAN-24	< 0.20	ug/L	No
Triallate	03-JAN-24	< 0.01	ug/L	No
Trichloroethylene	03-JAN-24	< 0.44	ug/L	No
2,4,6-Trichlorophenol	03-JAN-24	< 0.25	ug/L	No
Trifluralin	03-JAN-24	< 0.02	ug/L	No
Vinyl Chloride	03-JAN-24	< 0.17	ug/L	No



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Inorganic or Organic parameter(s) that have exceeded half the standard prescribed in Schedule 2 of Ontario Drinking Water Quality Standards:

During the reporting period the Running Annual Average for HAA's was exceeded during two quarters, Quarter 1 and Quarter 2.



2024 Summary Report to Council

In accordance with Schedule 22 requirements outlined in Ontario Regulation 170/03, a Summary Report shall be prepared no later than March 31 for the preceding year, and supplied to members of municipal council.

The Report shall list the requirements of the Act, Regulations, Drinking Water Works Permit, Municipal Drinking Water Licence, and any Orders applicable to the system that were not met at any time during the period covered by the Report.

The Report must also include a summary of the quantities and flow rates of potable water (water intended for human consumption), supplied during the Reporting Period, including monthly average and maximum daily flows. A comparison of these flows to the rated capacity and flow rates approved in the system Drinking Water Works Permit, and Municipal Drinking Water Licence, must also be provided.

Prescribed Instruments applicable to the Bayside DWS

The Bayside Service Area is governed by, and must operate their DWS primarily in accordance with, the following Acts and Regulations at minimum:

- Safe Drinking Water Act, 2002;
- O. Reg. 128/04 – Certification of Drinking Water System Operators and WQA
- O. Reg. 170/03 – Drinking Water Systems
- O. Reg. 169/03 – Ontario Drinking Water Quality Standards
- Environmental Protection Act, where applicable;
- Clean Water Act, where applicable;
- Municipal Drinking Water Licence 163-102;
- Drinking Water Works Permit 163-202;
- Permit to Take Water P-300-9303346859, expires December 6, 2034.



Councillors wishing to obtain a copy of any Act or Regulation are free to contact the Compliance Coordinator at collette.kingdon@quintewest.ca to obtain a current consolidated copy and interpretation of the legislation.

Compliance with Prescribed Instruments, Acts and Regulations

Safe Drinking Water Act

Adverse Water Quality Incidents as defined by the Safe Drinking Water Act have been listed in section [Notices submitted in accordance with subsection 18\(1\) of the Safe Drinking Water Act or section 16-4 of Schedule 16 of O.Reg. 170/03 and reported to Spills Action Centre during this Reporting Period](#). The following noteworthy activities occurred throughout the Reporting Period.

NSF International conducted a Surveillance Audit on November 21 and 22, 2024. The intent of the audit is to confirm continued conformance to the Drinking Water Quality Management Standard. The City received its Re-Accreditation, with no Non-Conformances identified in the Audit Report. Two (2) Opportunities for Improvement (OFI) were identified.

The City achieved a 100% Inspection Rating on its 2023 MECP Compliance Inspection conducted on October 4, 2023. No regulatory non-compliances were cited. At the time of writing this Report the MECP had not yet finished its 2024 Compliance Inspection, scheduled on January 29, 2025.

Clean Water Act

The Source Protection Plan was approved by the Minister of Environment, Conservation and Parks, and came into effect on January 1, 2015. The City has put the necessary internal processes in place with the Planning, Building, and Public Works Departments to screen



applications and ensure compliance with the Source Protection Plan. In 2024, three (3) applications were submitted to the Risk Management Official for review and approval surrounding the Bayside Intake Protection Zone. In general the screening, application, and approval processes implemented by staff seem to be working reasonably well. Throughout the screening and verification process it has been determined that currently there are no Risk Management Plans required for the properties identified surrounding the Bayside Intake.

Permit to Take Water

The City operates its Bayside Water Treatment Plant in accordance with Permit to Take Water (PTTW) number P-300-9303346859, which expires on December 6, 2034. This Permit allows the City to take 11,365 cu.m/day from the Bay of Quinte at a maximum flow rate of 262.8 L/s. The maximum recorded daily taking was 5,538 cu.m/day on November 14, 2024. The maximum recorded flow rate was 160.3 L/s on November 5, 2024.

For a detailed summary of water quantities and flow rates, see section [2024 Water Quantities and Flow Rates - Raw Water Assessment](#).

Drinking Water Works Permit/Municipal Drinking Water Licence

The Bayside Drinking Water System is now governed by Issue Number: 06 of Municipal Drinking Water Licence # 163-102, and Issue Number: 09 of Drinking Water Works Permit # 163-202. The Municipal Drinking Water Licence will expire September 28, 2025.

In accordance with Schedule C, Condition 1.1 of the MDWL, the Bayside Water Treatment Plant has a rated capacity to treat and distribute 11,360 cu.m/day. Over the Reporting Period the maximum daily volume of treated water distributed to the distribution system was 4,770 cu.m/day on December 9, 2024. The Annual Average Daily Flow pumped to the Distribution System was 3,412 cu.m/day. For a detailed summary of treated water



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quantities and flow rates, see section [2024 Water Quantity and Flow Rates \(Treated Water Assessment\)](#).

2024 Water Quantity and Flow Rates (Raw Water Assessment):

Raw Water Flow: PTTW Limit = 11,365 cu.m/d			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	3542	4370	121411
February	3855	4386	111789
March	3671	4215	113788
April	3489	5339	104684
May	3860	4556	119645
June	3975	4995	119236
July	4174	5003	129401
August	4112	5398	127479
September	3754	4388	112633
October	3578	4530	110910
November	3638	5538	109135
December	3963	4794	122843
Total Raw Water Flow =			1,402,952



2024 Water Quantity and Flow Rates (Treated Water Assessment):

Treated Water Flow: Facility Rated Capacity = 11,360 cu.m/day			
Month	Average Daily Flow (cu.m/d)	Maximum Daily Flow (cu.m/d)	Total Monthly Flow (cu.m/month)
January	3828	4458	118660
February	3778	4307	109561
March	3591	4099	111332
April	3388	4664	101646
May	3741	4409	115969
June	3795	4503	113843
July	4015	4693	124465
August	3829	4489	118699
September	3560	4510	107989
October	3474	4739	107689
November	3420	4032	102603
December	3902	4770	120950
Total Treated Water Flow =			1,353,406

- ➔ Annual Average Daily (Treated Water) Flow = 3,412 cu.m/day. This accounts for 30% of the facility Rated Capacity.
- ➔ Maximum Daily (Treated Water) Flow = 4,770 cu.m/day. This accounts for 42% of the facility Rated Capacity during recorded peak daily flow condition



Historical Flow Comparison

The 5-year ADF is calculated to be 3,491 cu.m/day, accounting for 31% of the facility rated capacity. This facility should still be able to distribute, on average, approximately 5,000 cu.m/day more water to meet average daily flow requirements.

During the 2024 calendar year the Bayside DWS supplied 412,040 cu.m of treated water to the Trenton DWS.

